

**An LLM Canary in the Online Data Coalmine: Bayesian Reasoning Problems as a
Capability-Gap Test for LLM Contamination in Online Samples**

Vera Wilde

Independent Researcher, Berlin, Germany

wilde.k.vera@gmail.com

Author Note

Vera Wilde <https://orcid.org/0000-0002-6470-4109>

Correspondence concerning this article should be addressed to Vera Wilde, Novalisstrasse 3,
10115 Berlin, Germany. Email: wilde.k.vera@gmail.com

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Abstract

The growing use of large language models (LLMs) by participants in online studies threatens the validity of behavioral research data. I propose that Bayesian reasoning problems, which have decades of well-replicated human performance benchmarks, can serve as calibrated detectors of LLM contamination in online samples — an approach I call a capability-gap test. In two preregistered pilots of a Bayesian reasoning training tool ($N = 148$), participants' accuracy on positive predictive value (PPV) calculation problems reached roughly three times established human performance ceilings. Specifically, 57% of Pilot 2 participants achieved perfect 5/5 scores, against a meta-analytic ceiling of approximately 24% for a single problem presented in natural frequency format, a level at which perfect scores are effectively unattainable. Including two outcome measures — accuracy (correct numerical answer) and Bayesian algorithm use (evidence of the reasoning process) — allows researchers to distinguish LLM contamination from authentic learning effects: accuracy detects contamination, while algorithm use preserves treatment effect signals. This is itself a signal detection problem, structurally analogous to the mass screenings for low-prevalence problems around which the Bayesian reasoning literature was developed. Bayesian reasoning problems offer four advantages as contamination detectors: well-established human performance ceilings from meta-analyses, a large human–LLM performance gap, known reference distributions enabling nuanced assessment, and ease of embedding in existing surveys. I provide practical recommendations for using these problems as data quality diagnostics in online research.

Keywords: capability-gap test, LLM contamination, data quality, online research, Bayesian reasoning, signal detection, natural frequencies, positive predictive value

An LLM Canary in the Online Data Coalmine: Bayesian Reasoning Problems as a Capability-Gap Test for LLM Contamination in Online Samples

The validity of online behavioral research data faces a growing threat: participants using large language models (LLMs) to complete study tasks. Platforms such as Prolific and Amazon Mechanical Turk have long been workhorses of behavioral research, offering rapid, affordable access to diverse samples. But the widespread availability of ChatGPT, Claude, and other LLMs since late 2022 means that any task requiring computation, reasoning, or text generation can now be outsourced to an AI assistant in seconds.

This creates a fundamental problem for researchers: how do you know whether your participants are thinking or outsourcing? Existing approaches to detecting automated or low-effort responding — such as completion time analysis, text pattern detection, CAPTCHAs, and attention checks — were designed for the pre-LLM era. They detect bots and inattentive humans, not humans strategically augmenting their performance with AI — saving time and potentially earning more per hour by completing tasks faster. Throughout, I use “LLM contamination” in the classical statistical sense of a contaminated sample — one in which an unknown fraction of observations is drawn from a different generating process (Tukey, 1960), here model-assisted rather than unaided responding — not in the machine-learning sense of benchmark data leaking into model training corpora (Dong et al., 2024). The problem is not hypothetical: shortly after ChatGPT's release, roughly a third of workers on Prolific — the platform used here — were estimated to have used an LLM on a text-production task (Veselovsky et al., 2025).

These are not the only tools available. A separate tradition uses statistical screeners, which — unlike direct screeners that flag individual responses (attention checks, completion-

time filters, CAPTCHAs) — ask whether a sample’s aggregate distribution is consistent with the target population (Arndt et al., 2022; DeSimone et al., 2015). Statistical screeners include pattern checks for straightlining and “Christmas-tree” responding, personal-reliability tests that re-administer items to check within-respondent consistency, and population-benchmark comparisons such as the “motorcycle test,” in which a sample’s rate on a characteristic with a known population value (e.g., motorcycle ownership) is compared against that value, with a large discrepancy taken as evidence that the sample is not drawn from the target population. My proposal belongs to this statistical-screener tradition but differs in a way that matters for the LLM era. Existing screeners target other failure modes: direct screeners catch bots and inattentive humans, pattern and reliability checks catch careless or inconsistent responding, and population-benchmark comparisons catch respondents who are not who they claim to be. None of them catches a competent, attentive respondent who is exactly who they claim to be — but augments performance with an LLM.

And, whereas a benchmark comparison like the motorcycle test detects non-representativeness — which a determined respondent could in principle fake by matching the known population rate — the Bayesian reasoning test detects a capability gap. I call this approach a capability-gap test (informally, an LLM canary): unaided human performance on natural-frequency positive predictive value (PPV) problems is reliably poor, so a sample that far exceeds the human ceiling reveals assistance that is hard to fake downward, since deliberately failing the problems defeats the purpose of using an LLM at all. Together, representativeness tests (of which the motorcycle test is one instance) and capability-gap tests span the space of population-benchmark screeners, and either kind can be constructed anew as older instances become known to survey takers.

I propose treating this as a signal detection problem in which we want to measure the LLM-associated noise and the remaining signal strength, and embedding Bayesian reasoning problems with known human performance ceilings as calibrated contamination detectors. This method exploits a unique property of these problems: the gap between human and LLM performance is large, well-documented, and mathematically precise.

This framing carries a structural irony worth making explicit at the outset, because it motivates the whole approach: detecting LLM contamination is itself a signal detection problem (Green & Swets, 1966); it shares an identical mathematical structure with the low-prevalence screening programs about which these Bayesian problems were designed to teach. The researcher screens a large sample for a rare-ish condition — unaided human reasoning — amid pervasive contamination. The unknown prevalence of unaided reasoning plays the role of the base rate. The known human performance ceiling plays the role of the test’s known error (that is, false positive and false negative) rates — fixing how often unaided humans produce an assisted-looking score. The gap between observed and expected performance is the test result, whose predictive value depends on both. This is the logic of the method.

Why Bayesian Reasoning Problems?

Consider a natural-frequency PPV problem. Of 1,000 people, 10 have a condition; of those 10, 8 test positive; of the 990 without it, 95 also test positive. If someone tests positive, what is the probability they have the condition? The answer is $8 / (8 + 95) \approx 8\%$, a figure most people find strikingly low. This is counter-intuitive due to the common cognitive bias known as base-rate neglect (Bar-Hillel, 1980; Kahneman & Tversky, 1973). Screening a large

population for something rare is searching for a needle in a haystack with a tool that throws up a second haystack — most of what it flags is not what was sought, and most people do not know that without some instruction in cognitive science and statistics. Thus, across three decades of studies, only about a quarter of participants solve such problems correctly, even in the facilitating natural-frequency format (McDowell & Jacobs, 2017). Current LLMs, by contrast, solve them all but perfectly. That large, stable gap is what makes these problems usable as detectors: when a sample's accuracy sits far above the human ceiling, the excess is a signature of LLM assistance rather than of unaided skill.

Bayesian inference problems — particularly those requiring calculation of positive predictive value (PPV) from base rates, sensitivity, and false-positive rates — have several properties that make them ideal contamination detectors:

Well-established human performance ceilings. Three decades of research has mapped the distribution of human performance on these problems across varied populations, formats, and task demands. Gigerenzer and Hoffrage (1995) established the seminal finding that presenting information in natural frequencies rather than conditional probabilities improves Bayesian reasoning. McDowell and Jacobs (2017) conducted a comprehensive meta-analysis establishing that the realistic average performance with natural frequency formats is approximately 24%, with 95% compatibility intervals spanning about 10–40% depending on specific conditions. Throughout, I use compatibility interval for what is conventionally called a confidence interval: the interval indexes the range of parameter values reasonably compatible with the data under the background assumptions, and the relabeling is intended to discourage the common misreadings that the interval expresses a degree of confidence in the values it contains, or that values outside it are incompatible with the data (Amrhein et al., 2019; Greenland et al., 2016). Weber et al. (2018) confirmed this ceiling and identified

“frequency phobia” as a contributing factor: people revert to probabilities to calculate PPV even when given frequency-format information. These benchmarks provide a well-calibrated reference distribution against which sample performance can be evaluated.

Large human–LLM performance gap. PPV calculations from natural frequency information are computationally straightforward — simple arithmetic involving addition and division. Current LLMs perform these calculations essentially perfectly. Human performance, even under optimal conditions (natural frequency format, educated samples), rarely exceeds 46% on individual problems (Gigerenzer & Hoffrage, 1995) and clusters around 24% in meta-analytic aggregation. This substantial gap between effectively perfect LLM performance and reliably poor human performance provides a strong contamination signal.

Known meta-analytic distributions. Rather than relying on a single-study benchmark, the reference distribution draws on meta-analytic estimates incorporating dozens of studies across populations and conditions. Researchers can compare their sample's performance not to a point estimate but to a well-characterized distribution, enabling nuanced contamination assessment.

Ease of embedding. A single PPV problem adds ~2–5 minutes to a survey and requires no specialized software. Problems can be drawn from published sets with known correct answers and known human performance benchmarks.

Method

Participants and recruitment. I conducted two OSF-preregistered online pilots (total $N = 148$) on Prolific to test Rarity Roulette, an interactive tool intended to help people understand mass screening for low-prevalence problems (e.g., polygraph testing, mammography), which share the mathematical structure of signal detection problems. Pilot 1

(December 2025) recruited 76 participants; Pilot 2 (January 2026) recruited 72. Both pilots preregistered a quality filter retaining participants who spent at least 15 minutes on task and passed effort and attention checks, yielding quality-filtered subsamples of 26 (Pilot 1) and 40 (Pilot 2). Participants were compensated at Prolific’s standard rate. Pilot 1 participants resided in the United Kingdom or United States; Pilot 2 recruitment was open to residents of the United Kingdom, United States, Ireland, Germany, Australia, Austria, New Zealand, or Switzerland. The Pilot 1 analysis sample comprises completed responses recorded after the study opened on Prolific; responses collected during pretesting are retained in the archived data but excluded from analysis.

Design and materials. Participants were randomly assigned to a treatment or control condition. The treatment group received the Rarity Roulette frequency-format learning intervention; the control group received a structured text explanation of the same problems in probability format (percentages, sensitivities, and false-positive rates, without counts or interactivity), modeled on standard medical risk-communication materials. Both conditions then solved five PPV calculation problems presented in natural-frequency format. For example: of 1,000 people, 10 have a condition; of those 10, 8 test positive; of the 990 without it, 95 also test positive; a person tests positive — what is the probability they have the condition? (correct answer $8 / (8 + 95) \approx 8\%$). The full instrument is available at <https://github.com/verawilde/rarity-roulette>. The pilots were designed to test the learning intervention; the contamination-detection application reported here emerged post hoc from unexpectedly high performance.

Measures and coding. Two outcome measures were collected for each problem. First, accuracy: whether the participant obtained the correct numerical answer (the secondary preregistered outcome). Second, Bayesian algorithm use: whether the participant’s “show

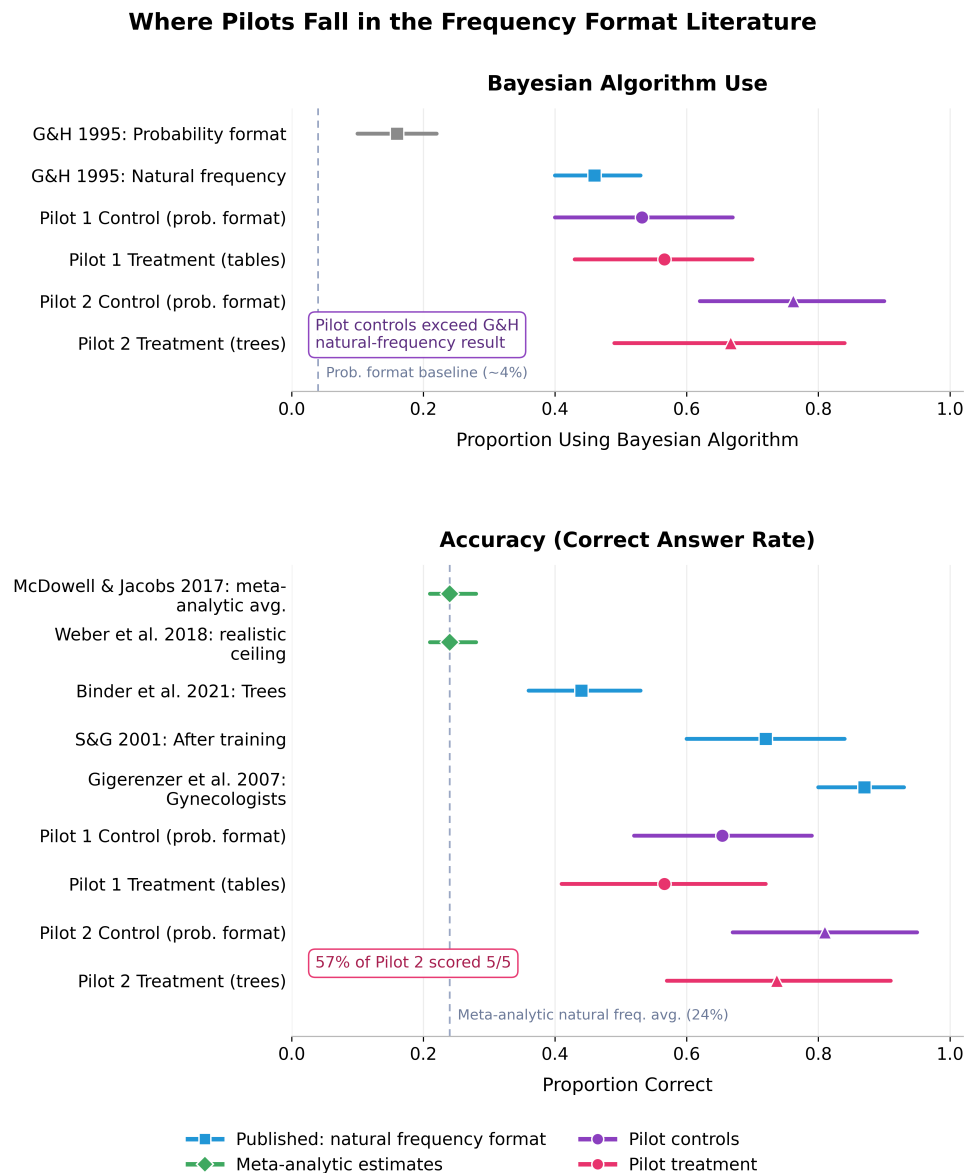
your work” response demonstrated Bayesian reasoning — identifying true and false positives and constructing the appropriate denominator (the primary preregistered outcome). Algorithm use was coded from free-text responses in Pilot 1 and from structured response fields plus free text in Pilot 2, following the rubric in the repository.

Analysis. I report proportions with 95% compatibility intervals, computed at the participant level, from each participant's proportion correct across the five problems, rather than by pooling the five items as independent observations. Responses within a participant are strongly dependent — participants tend to solve all five problems or none, which is what the concentration of perfect scores reported below reflects — so item-level pooling would understate uncertainty, and would do so in the direction that makes the gap above the human ceiling appear better determined than the data support. Since the argument of this paper is that the gap should be reported as a continuous diagnostic with its uncertainty attached, the wider and more conservative interval is the appropriate one. These are frequentist intervals throughout: although the substantive task concerns Bayesian reasoning, no Bayesian estimation was performed, and the intervals reported here are not credible intervals.

Treatment effects were assessed with a specification-curve analysis (Simonsohn et al., 2020). Rather than reporting a single estimate from one defensible set of analytic choices, a specification curve enumerates the reasonable alternatives — here, the alternative sample definitions — and displays the resulting estimates side by side, so that the sensitivity of the conclusion to those choices is visible rather than hidden in a single number. Across the specifications, the preregistered quality-filtered subsample yielded $d = -0.28$, 95% compatibility interval $[-0.91, +0.34]$. Preregistration, materials, data, and code are available at <https://github.com/verawilde/rarity-roulette> and <https://osf.io/exaqj>.

Results

Accuracy far exceeds established ceilings. Pilot 2 accuracy reflects the preregistered quality-filtered subsample; Pilot 1, which preregistered no time cutoff, is reported for the full sample. Sample sizes (N) denote the full recruited sample. Mean numbers of problems solved correctly, out of five, were as follows. Pilot 1 (December 2025; $N = 76$): control 3.27, treatment 2.83. Pilot 2 (January 2026; $N = 72$): control 4.05, treatment 3.68. Expressed as proportions correct, with 95% compatibility intervals: Pilot 1 control 65% [52, 79], treatment 57% [41, 72]; Pilot 2 control 81% [67, 95], treatment 74% [57, 91]. For transparency I also report the full Pilot 2 sample, which shows the same pattern at slightly lower levels (control 77% [65, 88], treatment 65% [51, 80]). Across the full Pilot 2 sample, 41 of 72 participants achieved perfect 5/5 scores (57%, 95% compatibility interval [45, 68]). Pilot 2 accuracy runs to roughly three times the meta-analytic human ceiling of approximately 24% correct for a single natural-frequency problem (McDowell & Jacobs, 2017): control accuracy is $3.2\text{--}3.4\times$ the ceiling and treatment $2.7\text{--}3.1\times$, depending on sample, while Pilot 1, though lower, still exceeds the ceiling by $2.4\text{--}2.7\times$. The gap is starker for perfect scores. At the ceiling rate, essentially no participant would be expected to solve all five problems correctly; 57% did. Control performance also exceeds the best-case trained-student benchmark of 46% (Gigerenzer & Hoffrage, 1995) and approaches the 87% achieved by gynecologists after a one-hour training session (Gigerenzer et al., 2007) — in an untrained Prolific sample (see Figure 1).

**Figure 1**

Pilot results in the context of published Bayesian reasoning benchmarks. Top panel: Bayesian algorithm use. Bottom panel: accuracy (correct answer rate). Error bars represent 95% compatibility intervals.

Bayesian algorithm use shows a more ambiguous pattern. Control group Bayesian algorithm use was 53% (Pilot 1; 95% compatibility interval [40, 67]) and 76% (Pilot 2; [62, 90]); treatment group algorithm use was 57% (Pilot 1; [43, 70]) and 67% (Pilot 2; [49, 84]).

These still exceed literature benchmarks, but the relative treatment effects (treatment minus control) fall within the range predicted by the meta-analytic literature. Specification curve analysis yields $d = -0.28$, 95% compatibility interval $[-0.91, +0.34]$ for the preregistered quality-filtered subsample, compatible with the realistic meta-analytic effect of $d \approx 0.35$ at the upper bound (see Figure 2). The relative treatment effect falls within the meta-analytic distribution, consistent with the view that apparent failures to replicate large effects may simply reflect draws from the lower end of authentic effect distributions (Amrhein et al., 2019).

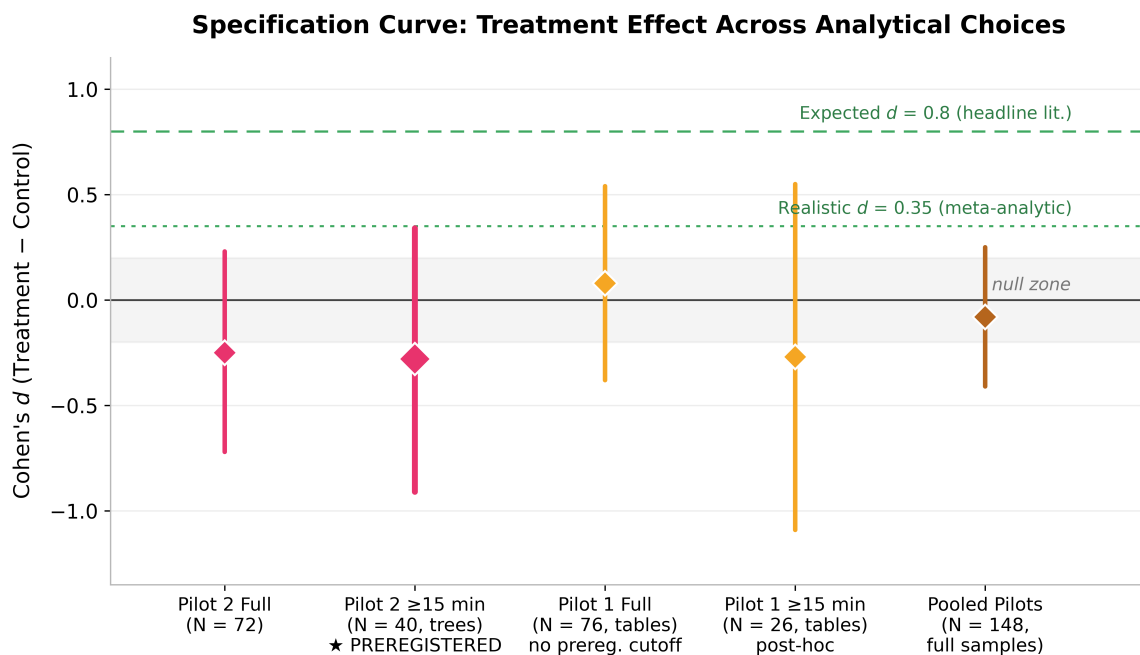


Figure 2

Specification curve analysis of treatment effects across analytical choices. The preregistered analysis (Pilot 2 ≥ 15 min) is highlighted. Dashed lines indicate expected ($d = 0.8$) and realistic meta-analytic ($d = 0.35$) effect sizes.

Discussion

The two pilots were not built to validate a contamination detector, but the pattern they produced constrains how such a detector should be specified and read. Three features of that pattern bear on the design of such a test: the divergence between the two outcome measures, the interpretability of the treatment–control difference, and the method's applicability beyond cognitive-task surveys. The divergent pattern across measures is informative. Accuracy — which requires only the correct number — shows the strongest contamination signal because LLMs produce correct numbers trivially. Bayesian algorithm use — which requires demonstrating reasoning in text — shows a more ambiguous pattern because LLM assistance may or may not generate text that resembles Bayesian reasoning, and because the relative treatment effect (the difference between conditions) can survive even when absolute levels are inflated. This divergence suggests a practical recommendation: researchers using Bayesian reasoning problems as LLM contamination detectors should collect both the numerical answer (for contamination detection) and a process measure (for preserving treatment effect signals). In expectation, the accuracy measure detects the contamination; the process measure preserves the experimental signal.

The treatment–control difference is itself interpretable within this framework. The manipulation delivered the frequency-format learning intervention to the treatment group; the resulting between-condition gap indexes that learning signal, and it survives in Bayesian algorithm use even as it is compressed by a ceiling in accuracy. Both conditions are inflated on accuracy because LLMs produce correct numbers trivially in either arm; that is why

accuracy carries the contamination signal, while algorithm use preserves the experimental one.

Why treatment accuracy runs below control in both pilots is not identifiable from these data. The compatibility intervals for the treatment effect include meaningful harm, meaningful benefit, and no effect, so the direction of the point estimates should not be over-read (Amrhein et al., 2019; Greenland et al., 2016). If the pattern reflects more than sampling variability, candidate explanations include friction from the interactive tool — the treatment arm showed higher dropout in both pilots — and the possibility that participants who had just worked through the intervention attempted the problems unaided while control participants outsourced them, in which case the negative point estimate would itself be partly a contamination artifact. The present design cannot distinguish these accounts.

The method's requirement of a correct answer might seem to limit it to cognitive-task surveys. It does not. Personality and other non-cognitive instruments lack a right answer and so cannot self-diagnose contamination, but one or two PPV problems can be embedded alongside them purely as a diagnostic, yielding a contamination estimate for that data collection without altering the substantive items. In this role the Bayesian problems function as a low-cost LLM canary in surveys that otherwise offer no internal check on LLM use.

Practical Recommendations

First, embed one to two PPV problems with known correct answers from the published literature into online surveys. Problems should present information in natural frequency format to maximize the diagnostic value of the known ceiling.

Second, collect two outcome measures: accuracy (correct numerical answer) and a process measure (free-text explanation or structured reasoning indicators). This allows simultaneous contamination assessment and treatment effect preservation.

Third, compare sample performance to the benchmarks introduced above — the ~24% meta-analytic average (10–40% compatibility interval) and Gigerenzer and Hoffrage’s (1995) 46% best-case student result. The further a sample’s performance falls above the upper bound of these distributions, the stronger the evidence for contamination. The magnitude and uncertainty of the gap should be reported as a continuous diagnostic, not reduced to a binary contamination/no-contamination decision.

Fourth, report contamination diagnostics. Even if PPV problems are not the study’s focus, reporting the performance gap as a data quality diagnostic helps the field calibrate expectations for online data collected in the LLM era.

Fifth, do not use these diagnostics to penalize individual participants. The method provides sample-level contamination estimates, not individual classifications — using it to withhold pay or reject submissions would create the very false-positive problem these questions illustrate. Preregistered subgroup analyses comparing participants who exceed literature benchmarks to those within expected ranges may be informative, but researchers should report both subgroups’ results rather than simply discarding data.

Limitations

Several caveats apply. First, the 10–40% meta-analytic benchmark range comes primarily from Western, Educated, Industrialized, Rich, and Democratic (WEIRD) samples (Henrich et al., 2010); cross-cultural variation in Bayesian reasoning exists (Wu & Emery,

2021) but is less well-characterized, which limits the generalizability of these benchmarks as contamination detectors in non-WEIRD populations.

Second, Bayesian algorithm use was coded by an automated keyword-and-pattern rule rather than by human raters. That rule detects the vocabulary and arithmetic of the Bayesian algorithm rather than verifying that it was applied correctly, so the algorithm-use rates reported here should be read as upper bounds. This is a source of uncertainty distinct from contamination, and it cuts in the opposite direction from the paper's argument: an overinclusive coder inflates apparent algorithm use and would tend to overstate rather than understate the excess above literature benchmarks. Accuracy, the measure carrying the contamination signal, is scored on exact numerator-and-denominator matching and is not subject to this concern.

Third, participants may use LLMs for some tasks and not others within a single survey. Embedding PPV problems provides a contamination estimate for those specific items; contamination levels on other survey items may differ in either direction. Because contamination may be item-specific, the strongest inference comes from combining detectors rather than relying on any one. The capability-gap test is complementary to existing aberrant-response methods — straightlining and Christmas-tree checks, personal-reliability tests, and population-benchmark comparisons — each of which targets a different failure mode. Straightlining checks catch inattentive patterning; personal-reliability tests catch inconsistency; the capability-gap test catches competent LLM assistance the others miss. Deployed together, they yield converging evidence about the presence and location of LLM use within a survey.

Fourth, contamination rates will likely increase over time as LLMs become more integrated into everyday tool use, which will shift what “uncontaminated” performance looks

like. Indeed, the descriptive statistics from my two pilots may already illustrate this trend: Pilot 2 (January 2026) showed substantially higher accuracy than Pilot 1 (December 2025) across both conditions, consistent with increasing LLM adoption over even a short interval. Fifth, my pilots were designed to test a learning intervention, not to validate a contamination detection method; accordingly, the contamination interpretation offered here is post hoc and suggestive rather than confirmatory. A dedicated validation study with controlled LLM access conditions would be needed to establish the sensitivity and specificity of PPV problems as contamination detectors. Sixth, the signal detection analogy has limits — unlike mass screening, we have some control over the “base rate” of contamination through study design choices.

Conclusion

The golden age of online data collection — to the extent it ever was one (Chmielewski & Kucker, 2020) — may be over, but the data quality problem it leaves behind can be partially addressed with tools from the very literature that documented human reasoning limitations in the first place. Bayesian reasoning problems provide cheap, well-calibrated, easy-to-embed capability-gap tests for LLM contamination in online research. Having both accuracy and process measures allows researchers to detect the contamination while preserving an experimental signal amid the noise. Ironically, this is itself a (constructed) signal detection problem — using knowledge about the ceiling of unaided human performance to screen samples for the rare condition of unassisted thought.

Open Practices Statement

Both pilot studies were preregistered on OSF (<https://osf.io/exaqj>). All materials, data, and analysis code are available at <https://github.com/verawilde/rarity-roulette>. Neither study was designed to validate the contamination detection method proposed here; the proposal emerged from post hoc analysis of unexpected results.

Declarations

Funding: No funding was received for this study.

Competing interests: The author has no relevant financial or non-financial interests to disclose.

Ethics approval: Both studies were conducted on the Prolific platform with informed consent.

This research was conducted independently without institutional affiliation; accordingly, no institutional review board reviewed the protocol. Both studies followed Helsinki Declaration principles and Prolific's participant protection standards.

Consent to participate: Informed consent was obtained from all participants via the Prolific platform.

Consent for publication: Not applicable.

Data availability: All data and analysis code are available at <https://github.com/verawilde/rarity-roulette>. Preregistrations are available at <https://osf.io/exaqj>.

Code availability: Analysis code is available at <https://github.com/verawilde/rarity-roulette/tree/main/analysis>

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